



General Use SOP

Highly Acutely Toxic Materials

Process or Experiment Description

This standard operating procedure (SOP) provides general guidance on working safely with lab-scale quantities of highly acutely toxic materials. In some cases, multiple SOPs may apply to a single chemical (e.g., for sodium azide, both General Use SOPs for highly acutely toxic and highly reactive/unstable materials could apply). For questions about the applicability of any item in this SOP, contact your Principal Investigator/Supervisor or EH&S at 650-723-0448.

For work with larger quantities of highly acutely toxic materials, complex procedures, or otherwise particularly hazardous procedures involving carcinogenic materials, EH&S recommends completing a [risk assessment](#), reviewing other EH&S guidance (such as [guidance on scale-up](#)), looking for SOPs on similar procedures in the [SOP Library](#), and/or writing a [dedicated SOP](#) and [submitting it for EH&S review](#).

Hazards

Highly acutely toxic substances have the ability to cause severe adverse health effects, including death, after a single exposure/dose or multiple exposures/doses within a 24 hour period.

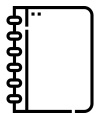
A material is considered highly acutely toxic if any of the following hazard statements are listed in the material's SDS:

- H300: Fatal if swallowed
- H310: Fatal in contact with skin
- H330: Fatal if inhaled

In general, the lower a compound's LD50 or LC50 is, the more toxic it is. The route of exposure impacts on the associated risk. For example, potential dermal or inhalation exposures are more common in workplaces than potential ingestion exposures. Mixtures may also affect the risk of exposure. For example, dimethylsulfoxide (DMSO) is not toxic itself, but it can carry toxic materials dissolved in it through the skin, even if the toxic material is not normally toxic by dermal exposure.

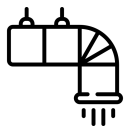
Common highly acutely toxic substances include sodium azide, potassium cyanide, [hydrofluoric acid \(HF\)](#), [tetramethylammonium hydroxide \(TMAH\)](#), and osmium tetroxide.

Control of Hazards



General

- Minimize the potential for splash, splatter, or other likely scenarios for accidental contact.
- Consider using a less toxic material if feasible.
- Use the smallest quantity and concentration feasible.
- Minimize the potential for splash, splatter, or other likely scenarios for accidental contact.
- Work with highly acutely toxic materials only when there is someone else in the lab ready to assist you in the event of an incident.
- Weigh solids in a chemical fume hood or ducted balance enclosure.
- The tare method can prevent chemical inhalation while keeping the balance on the bench.
 - In a chemical fume hood, add the chemical to a preweighed container.
 - Seal the container and reweigh it outside of the hood.
 - Adjust the amount by adding or removing the chemical inside the hood.
- Be diligent about lab hygiene - avoid touching your face or items that you will handle with ungloved hands (e.g., a smartphone). Change gloves regularly and thoroughly wash your hands immediately after completing work with the material.



Engineering/Ventilation Controls

Use a properly functioning [chemical fume hood](#) when handling highly acutely toxic materials.

If the process doesn't allow working in a fume hood, contact Environmental Health and Safety at 650-723-0448.



Personal Protective Equipment

In addition to proper street clothing (long pants or equivalent that covers legs and ankles, and close-toed non-perforated shoes that completely cover the feet), wear the following Personal Protective Equipment (PPE):

- Safety glasses (if splash potential exists, wear goggles and a face shield instead)
- Lab coat
- Appropriate [chemical-resistant gloves](#)



Special Handling Procedures and Storage Requirements

Containers should be closed when not in use. [Segregate](#) incompatible chemicals and use [secondary containment](#). Follow any substance-specific storage guidance provided in SDS.

Emergency Procedures

All incidents should be reported to EH&S with an incident report form.

Spill

Refer to EH&S' guidance on hazardous materials incidents for general procedures for incident response, and call 650-725-9999 for assistance with spill cleanup.

If you are trained in spill cleanup and the spill is not health-threatening, proceed with spill cleanup. There may be variations in procedure based on the specific spill, but the general steps are:

1. Don the appropriate PPE (See Personal Protective Equipment section, above).
2. Locate the nearest spill kit and bring it to the spill.
3. Use the absorbent pads in the spill kit to wipe up the spill.
4. Place the used absorbent pad and gloves inside the spill kit jar.
5. Place a waste tag on the spill kit and request waste pickup.

Exposure

If immediate medical attention is required, call 911. Remove any contaminated clothing, and IMMEDIATELY flush contaminated skin with water for at least 15 minutes. Use a safety shower for any skin exposures to the trunk, legs, or feet.

For eye exposures, IMMEDIATELY flush eyes with water for at least 15 minutes using an eyewash station.

If medical attention is needed, provide the SDS(s) of the chemical(s) to aid medical staff in proper diagnosis and treatment.

For minor exposures, personnel can contact the Stanford University Occupational Health Center at

650-725-5308. For serious exposures, personnel should go to the Stanford Hospital Emergency Department.

Those who work with hazardous chemicals are entitled to receive medical attention/consultation when:

- A spill, leak, explosion or other occurrence results in a hazardous exposure (potential overexposure).
- Symptoms or signs of exposure to a hazardous chemical develop.

Waste Disposal

Dispose of highly acutely toxic materials as hazardous waste. All hazardous waste containers should have a waste tag attached before any waste is added. Before combining materials in waste containers, review the SDSs and Stanford's [Incompatibility Guide](#). For general guidance regarding waste disposal, refer to the guide for [handling and storing waste](#) and the [chemical waste poster](#).

Minimum Training Requirements

- General Safety & Emergency Preparedness (EHS-4200)
- Chemical Safety for Laboratories (EHS-1900)
- Laboratory-specific training

Approval Required

Consult with your PI regarding the need for prior approval. Laboratory personnel shall seek and the PI must provide prior approval of any chemical usage involving [Restricted Chemicals or High Risk Procedures](#). EH&S recommends obtaining prior PI approval for work with highly reactive/unstable materials.

Designated Area

For use of highly toxic materials, a designated area shall be established where limited access, special procedures, knowledge, and work skills are required. A designated area can be the entire laboratory, a specific laboratory workbench, or a laboratory hood.

Designated areas must be clearly marked with signs that identify the chemical hazard and include an appropriate warning; for example: WARNING! HYDROFLUORIC ACID WORK AREA – HIGHLY TOXIC MATERIAL.

- Upon leaving the designated area, remove any personal protective equipment worn and wash hands, forearms, face, and neck.
- After each use (or day), wipe down the immediate work area and equipment to prevent accumulation of chemical residue.
- At the end of each project, thoroughly decontaminate the designated area before resuming normal laboratory work in the area.